1. This question relates to the design of an instrument for measuring bio-potentials from neurons in the brain.

   a) Explain why a potential of 70mV exists across the cell membrane of a neuron when it is at rest. [3]

   b) Figure 1.1 shows an AC coupled bio-potential amplifier for measuring neural activity from the brain. The electrodes used have a capacitance C_E=60μF and V_{sig} and V_{os} denote the measured bio-potential and common mode offset.

      i. Give one advantage and one disadvantage of using AC rather than DC coupled bio-potential amplifiers for this application. [2]

      ii. C_1 and R_1 are used to AC couple the input signal to the amplifier. C_1 is chosen to be 100pF. State why 100pF is a suitable value. [1]

      iii. Derive the input voltage to the amplifier V_{in}, and show that it is high pass function stating the equation of the cut off frequency stating any assumptions made. To aid your calculations you may use just one branch of the differential input for your analysis. [2]

      iv. Calculate a suitable value for R_1 to give you a high pass cut off at 10 Hz. [1]

   c) The Bio-potential amplifier in Figure 1.1 is used as part of a monitoring system to detect the action potential signal shown in Figure 1.2. The gain of the bio-potential amplifier is set to A=1000. The output of the Bio-potential amplifier is band-pass filtered between 0.3KHz and 5KHz with a 20dB/decade high-pass followed 40dB/decade low-pass filter. A final gain stage is added to maximize the dynamic range between +3V to -3V.

      i. Sketch the complete schematic of this monitoring system, calculating suitable resistor values for your filters. You may assume all capacitors are, C=100pF for your filters. [5]

      ii. Choose suitable gains and calculate resistor values to ensure that the action potential shown in Figure 1.1 utilises the maximum dynamic range of the measurement system. [4]

   d) The bio-potential amplifier shown in Figure 1.1 is to be modified to allow a fully integrated implementation in CMOS for recording from a micro-needle array. The requirements are the gain of the modified amplifier is set by a ratio of capacitors and the output and input common mode voltages at DC are the same. Sketch the schematic of the modified bio-potential amplifier. [2]

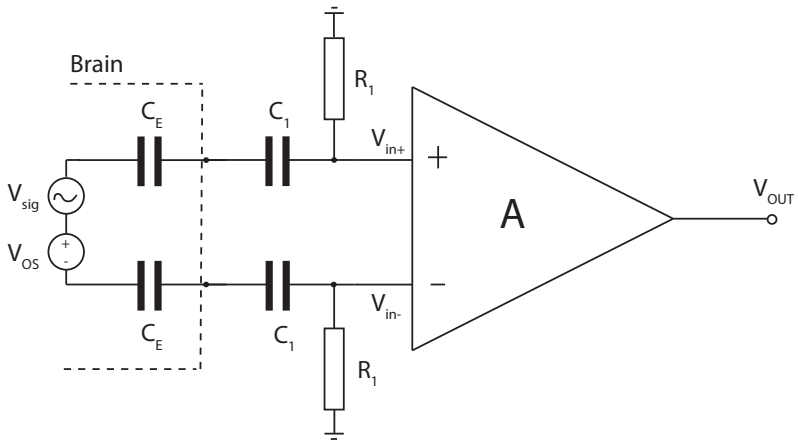


Figure 1.1

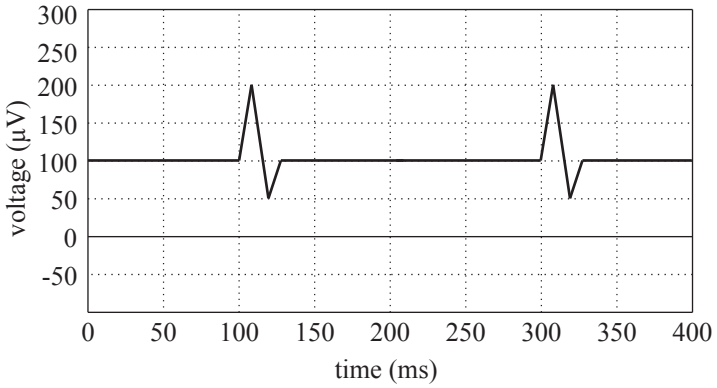


Figure 1.2

2. This question relates to a 2-channel *electrical stimulation circuit for a peripheral nerve interface*.

a) Describe the main benefit and also challenge in the design and implementation of: (i) current-mode; (ii) voltage-mode; (iii) charge-mode electrical stimulation circuits. [6]

b) Sketch a charge-balanced, biphasic stimulation waveform profile to achieve the following parameters: [4]

Electrode configuration:	Bipolar
Stimulus magnitude:	500nC
Repetition rate:	100Hz per channel
Stimulation strategy:	Continuous interleave sampling

c) Determine the conductor geometry for an appropriate cuff electrode to deliver this stimulus to a nerve bundle of diameter 0.5mm considering a platinum iridium conductor (with maximum charge injection capacity of $500\mu C/cm^2$). [4]

d) Design a circuit for generating the stimulation waveform profile in your answer to (b). [6]

3. This question relates to the design of chemical sensor instrumentation for DNA detection.

a) Explain the principles of cyclic voltammetry using illustrations where necessary.

[4]

b) Figure 3.1 shows an amperometric sensor used for the detection of DNA hybridization. The sensor works on the principle of a redox reaction confirming hybridization on WE1 at a cell potential of V_{cell}=100mV. The maximum expected output current during hybridization is 10nA. In the same time the sensor drift is expected to be 1uA. An additional electrode, WE2 is added for sensor drift compensation. It is desirable that the output of any instrumentation system use the difference between the hybridization reaction and drift to confirm a reaction.

i. Design instrumentation to interface to this sensor which provides the cell bias, V_{cell} and a suitable readout which can utilize the second electrode, WE2 to increase the signal-to-noise ratio at the output. You are required to show all your schematics and label voltages where necessary and calculate values for all components used stating assumptions where necessary. You may assume any OpAmps used operate between 0-1.8V and you are required to use the full dynamic range of your system.

[8]

ii. Explain why the counter electrode, CE, is larger than the working electrode, WE1.

[2]

c) Figure 3.2 shows an Ion-Sensitive Field Effect Transistor.

i. State two reasons why this sensor better suited for DNA detection than the one shown in Figure 3.1.

[2]

ii. The sensor shown in Figure 3.2 is to be used in a DNA detection platform which also includes a Reference-FET (REFET) for background subtraction of any drift. Sketch the schematic of suitable circuit which allows differential measurement from the ISFET and the REFET.

[4]

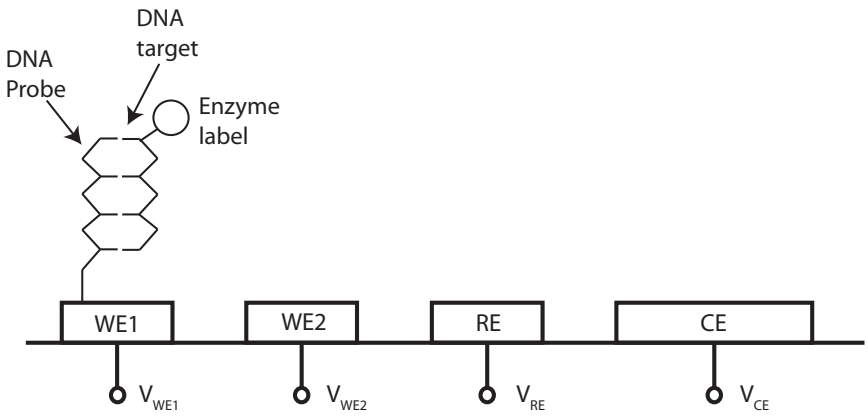


Figure 3.1

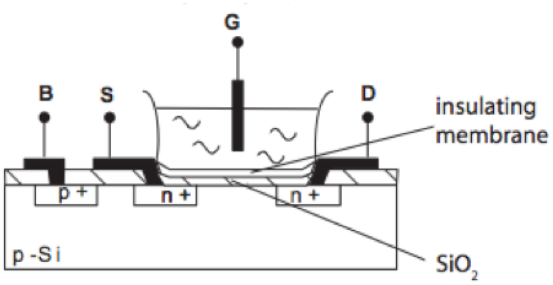


Figure 3.2

4. This question concerns alternative power supply methods for body worn and implanted medical electronics.

a) Explain, with the aid of a diagram, the structure and principle of operation of a thermo-electric generator (TEG).

[4]

b) A TEG is attached the human body with the aim of powering a wearable device in an ambient temperature that ranges from -2ºC to 30ºC.

i) Draw and explain an electrical equivalent circuit model to represent the flow of thermal energy through the system from the core of the body to the ambient.

[4]

ii) Assuming the TEG is 100% efficient at converting heat flow into electrical power, calculate the upper limit on the useful electrical output power that can be obtained for a TEG when the thermal resistance from body-core to TEG is 200 K/W, and the thermal resistance from TEG to ambient is 20 K/W.

[5]

c) Motion-driven energy harvesters are candidates for powering medical electronics. Explain, with the aid of a diagram, two ways in which an electrostatic transducer can be operated to convert mechanical energy to an electrical form.

[4]

d) Explain the similarities and differences between a regular cored-transformer and an open-cored inductive power transfer system

[3]

5. This question concerns diabetes management systems.

a) Describe three essential components needed to develop an artificial pancreas (i.e. closed-loop insulin delivery system) and draw the corresponding closed loop block diagram.

[5]

b) State four reasons why an artificial pancreas is still not yet able to achieve optimal blood glucose control (normoglycemia) with the current available technology.

[4]

c) Explain the function of an insulin bolus calculator and state its formula explaining all terms.

[4]

d) Using the insulin bolus calculator from part c), compute the insulin dose required for the following scenario:
   - Amount of carbohydrate eaten: 70 grams
   - Insulin to carbohydrate ratio: 10 gram/unit
   - Current blood glucose: 10 mmol/L
   - Target blood glucose: 6 mmol/L
   - Insulin sensitivity factor: 2 mmol/L
   - Insulin on board: 1 unit

[3]

e) Sketch a block level diagram of a glucometer that could be used to measure blood glucose in a person with diabetes.

[4]